

A chlorinated organic cation enables stable 2D/3D tin iodide perovskite photovoltaics

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Tin halide perovskites (THPs) offer narrower bandgaps and improved environmental safety compared with the widely studied lead-based perovskites, but their air sensitivity has hindered their progress and demands new material design to enable their practical applications. Here we report stable and efficient 2D/3D tin perovskite solar cells enabled by new ultrastable 2D and quasi-2D THPs based on the 4-chloro-phenethylammonium (4CIPEA) cation. The stronger π -stacking interactions and tighter interlayer packing in $(4\text{CIPEA})_2\text{SnI}_4$ among $(4\text{XPEA})_2\text{SnI}_4$ structures ($X = \text{H, F, Cl, Br}$) substantially impede oxygen and water diffusion, enabling superior air and moisture stability and bright photoluminescence lasting several months in ambient air. The addition of 4CIPEA markedly improves 2D/3D THP film crystallinity and orientation, leading to 16.2% efficient 2D/3D THP solar cells that show prolonged storage stability and operational stability at 55 °C surpassing 1,000 h. This study establishes a new strategy for designing stable and efficient THPs towards their practical applications.

Over the past decade, metal halide perovskite solar cells (PSCs) have rapidly progressed as highly tunable, solution-processable thin film photovoltaics that now yield single-junction power-conversion efficiencies (PCEs) comparable to state-of-the-art silicon solar cells. The current champion PSCs have attained over 27% PCE and are based on lead perovskites with intermediate 1.5–1.6-eV bandgaps^{1–3}, but the high toxicity of lead poses potential biological and environmental hazards⁴. These toxicity concerns have stimulated the exploration of lead-free halide perovskites, among which tin halide perovskites (THPs) are the most promising for their higher carrier mobilities, robust excitons and narrower 1.3–1.45-eV bandgaps within the ideal range for

the Shockley–Queisser radiative limit^{4–10}. However, compared with lead perovskites, THPs have greater Lewis acidity, rapid and less controlled crystallizations and greater susceptibility of oxidation from Sn^{2+} to Sn^{4+} (refs. 11–13). Consequently, pure tin-based PSCs have only recently surpassed 17% PCE through elaborate transport layer and film crystallization engineering^{14–16}. Beyond improvements to efficiency, the inherent stability of the THP active layers must be substantially improved to enable their practical applications^{17–20}.

To improve the crystallization and stability of 3D $(\text{A})\text{SnI}_3$, large organic ‘spacer cations’ that can form layered 2D and quasi-2D $(\text{LA})_2(\text{A})_{n-1}\text{Sn}_n\text{I}_{3n+1}$ perovskites are commonly used (where LA is the

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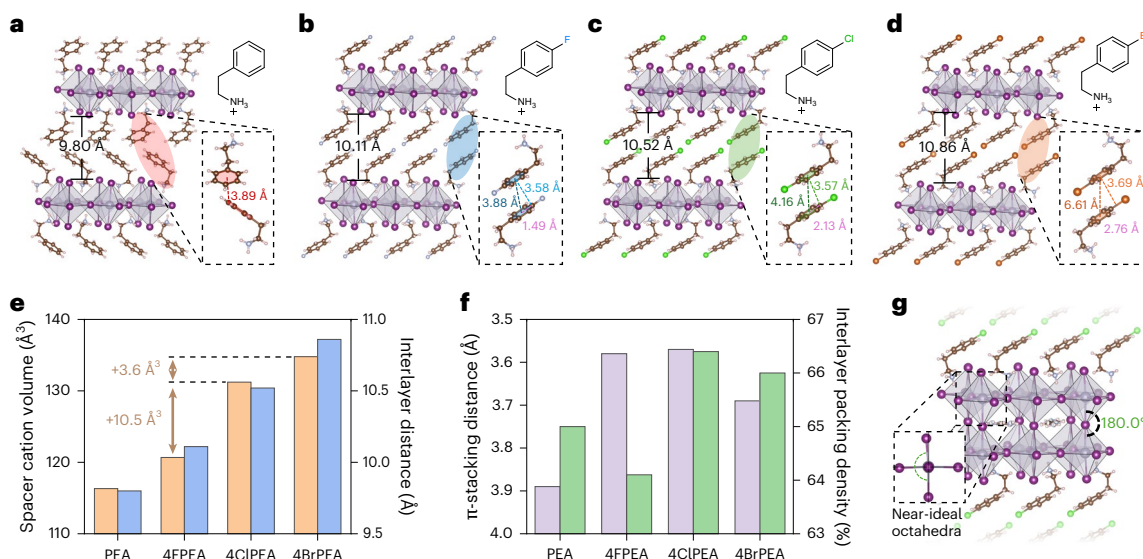


Fig. 1 | Crystal structural analysis of 4XPEA-based 2D THPs (X = H, F, Cl, Br). **a–d**, Crystal structures of $(\text{PEA})_2\text{SnI}_4$ (ref. 49) (**a**), $(4\text{FPEA})_2\text{SnI}_4$ (ref. 34) (**b**), $(4\text{ClPEA})_2\text{SnI}_4$ (**c**) and $(4\text{BrPEA})_2\text{SnI}_4$ (**d**) with their corresponding spacer cation structures and insets highlighting the respective interlayer π – π interactions between XPEA. Minor disorder components are omitted for clarity. **e**, Spacer cation volumes (orange) and the interlayer (terminal I–I) distances (blue) in the

$n = 1$ $(4\text{XPEA})_2\text{SnI}_4$ structures. **f**, π –stacking distances (lavender) and interlayer packing density (green) for the different $(4\text{XPEA})_2\text{SnI}_4$ structures. The π –stacking distances are defined by the carbon-to-centroid distance in $(\text{PEA})_2\text{SnI}_4$ and aromatic plane-to-plane distances in others. **g**, Crystal structure of $n = 2$ $(4\text{ClPEA})_2(\text{FA})\text{Sn}_2\text{I}_7$, highlighting minimized intraoctahedral and interoctahedral distortions and an ideal 180° out-of-plane Sn–I–Sn angle.

spacer cation and A is a small monovalent cation such as Cs^+ , methylammonium (MA) or formamidinium (FA))^{17–19,21,22}. The spacer cations in mixed 2D/3D PSCs not only improve film crystallinity and orientation owing to their templating effects^{18,22}, but also enhance air stability over 3D PSCs owing to their hydrophobic character and grain passivation capability^{23,24}. Furthermore, the spacer cations can modulate the perovskite structural distortions and photophysical properties, including the bandgap and quantum yield^{25–28}. Most recent champion tin PSCs have adopted 2D/3D compositions using the prototypical phenethylammonium (PEA) spacer cation that provides high quantum yield, low electron–phonon coupling and improved air stability over aliphatic spacers such as butylammonium^{27,29,30}. However, even PEA has limited protective capabilities in THPs^{31,32}. To achieve more stable and efficient 2D/3D tin PSCs, the spacer cations must (1) show increased rigidity, hydrophobicity and tight interlayer packing to enhance air stability³²; (2) impart minimal distortion onto the tin iodide lattice to ensure optimal charge transport; and (3) template the crystal growth of 3D THPs effectively with 2D or quasi-2D structures to inhibit degradation and enhance 2D/3D film crystallization.

In this Article, to overcome these challenges and develop more air-stable 2D THPs for 2D/3D tin PSCs, we study selective halogenation of the spacer cation, inspired in part by recent reports showing that chlorinated electron-withdrawing surface passivators (which do not form 2D perovskites) can improve PSC surface passivation and performance^{1,33}. Using PEA as a baseline, we focus on para-substituted 4XPEA cations (X = H, F, Cl, Br) owing to their minimal distortion on perovskite lattices compared with ortho- and meta-substituted analogues³⁴. One such 4XPEA spacer cation, namely 4-fluorophenethylamine (4FPEA), has been shown to improve the crystallinity and stability of Pb ^{35–38} and Sn ^{39,40} PSCs beyond PEA. It has been generally assumed that spacer cation fluorination leads to the highest stability owing to the high electronegativity of fluorine and resultant hydrophobicity. However, tin perovskites using 4XPEA cations with halogens beyond fluorine have rarely been studied⁴¹, lacking crystal structure reports and deeper insights connecting 4XPEA functionalization with structure, crystallinity and, most importantly, stability outcomes for THPs. Notably, chlorine has the highest electron

affinity and a larger atomic radius than fluorine, which can lead to distinct supramolecular halogen– π interactions^{42,43} that could influence THP growth and stability. These considerations and the need for systematic comparisons motivated us to investigate the influence of spacer cation halogenation (X = F, Cl, Br) towards the goal of improving tin 2D/3D PSCs.

Herein, we report new 2D (and quasi-2D) THP crystal structures based on the 4-chloro-phenethylammonium (4ClPEA) and 4-bromo-phenethylammonium (4BrPEA) spacer cations and systematically compare the capabilities of para-functionalized 4XPEA spacer cations in regulating structural distortion, air stability and film crystallinity of tin PSCs. We found that $(4\text{ClPEA})_2\text{SnI}_4$ shows minimal octahedral structural distortions, strong π – π stacking interactions and the tightest interlayer packing compared with other $(4\text{XPEA})_2\text{SnI}_4$ analogues. Density functional theory (DFT) calculations reveal that 4ClPEA uniquely blocks major O_2 and H_2O diffusion paths across the interlayer, yielding the highest minimum diffusion barriers and suppressing O_2 and H_2O diffusion rates in $(4\text{ClPEA})_2\text{SnI}_4$ by many orders of magnitude relative to those in $(\text{PEA})_2\text{SnI}_4$. These favourable characteristics enable ultrastable 2D THPs that maintain high crystallinity and bright red photoluminescence over several months in ambient air. The addition of 4ClPEA salt into 3D (FA) SnI_3 inks also substantially improves the film crystallinity, orientation and surface morphology over the other 4XPEA salts. Owing to these improvements, we demonstrate champion 4ClPEA-based 2D/3D tin PSCs with $\sim 16.2\%$ that retain over 95% of their initial efficiency after 1,600 h in dry air and 80% of their initial efficiency after 1,000 h under continuous 1-sun illumination at 55°C in air, marking a major step towards their practical applications.

Influence of 4XPEA on THP crystal structures

We synthesized single crystals of new $n = 1$ $(4\text{ClPEA})_2\text{SnI}_4$, $(4\text{BrPEA})_2\text{SnI}_4$ and $n = 2$ $(4\text{ClPEA})_2(\text{FA})\text{Sn}_2\text{I}_7$ THPs via slow cooling crystallization and determined their crystal structures using single-crystal X-ray diffraction (XRD) (Supplementary Table 1). We also synthesized an analogous $(4\text{ClPEA})_2(\text{MA})\text{Sn}_2\text{I}_7$ structure (Supplementary Fig. 1) but focus on the $n = 2$ FA analogue owing to the greater relevance of FA in tin PSCs^{14,15,39,40}. To establish the general effects of these spacer cations, we compared the

structural distortions of the $(4\text{XPEA})_2\text{SnI}_4$ phases (Fig. 1a–d). Each structure shows comparably low intraoctahedral distortions, as described by the similar bond angle variance (σ_1^2), metal off-centring distortion (σ_2^2) and bond distortion index D values (Supplementary Table 2 and Note 1)¹². Meanwhile, minor differences are found in the interoctahedral Sn–I–Sn bond angles: 155.72°, 156.37°, 156.97° and 157.01° for $(\text{PEA})_2\text{SnI}_4$, $(4\text{FPEA})_2\text{SnI}_4$, $(4\text{CIPEA})_2\text{SnI}_4$ and $(4\text{BrPEA})_2\text{SnI}_4$, respectively. These differences translate to slight redshifts in photoluminescence (PL) emission wavelengths (Supplementary Fig. 2), wherein the more optimal bond angles improve orbital overlaps and thus produce narrower bandgaps²⁶.

Profound differences are observed in the organic interlayers of the four $(4\text{XPEA})_2\text{SnI}_4$ structures. Similar to the well-studied $(\text{PEA})_2\text{PbI}_4$ (ref. 35), the PEA cations in $(\text{PEA})_2\text{SnI}_4$ adopt indirect edge-to-face π – π interactions (Fig. 1a, inset). However, para-substitution with halogens (X = F, Cl, Br) leads to stronger face-to-face parallel slip-stacked π – π interactions between the 4XPEA dimers in $(4\text{XPEA})_2\text{SnI}_4$ (Fig. 1b–d, insets). This face-to-face stacking confers greater rigidity of the organic interlayer and stronger templating behaviour^{35,40}. Comparable aromatic plane-to-plane distances of ~3.57 Å are found between the strongly interacting 4XPEA dimers in $(4\text{FPEA})_2\text{SnI}_4$ and $(4\text{CIPEA})_2\text{SnI}_4$; however, a slightly larger distance of 3.69 Å is found in $(4\text{BrPEA})_2\text{SnI}_4$ owing to the much larger atomic radius of Br. The different substituents also lead to notably different ring centroid-to-centroid distances (3.88 Å, 4.16 Å, 6.61 Å) and displacements (1.49 Å, 2.13 Å, 2.76 Å) between 4FPEA, 4CIPEA and 4BrPEA dimers, respectively, to accommodate the increasingly larger atomic radii of the halogen substituents. Importantly, Cl functionalization offers the largest benefits towards increasing the spacer cation molecular volume without dramatically expanding the π -interaction distances between cations or the interlayer distance between perovskite layers (Fig. 1e). In addition, the Cl substituent may also enhance supramolecular donor to π -acceptor interactions compared with F^{42–44}, and thus could further strengthen the interactions between 4CIPEA units.

Beyond these critical differences in the π -stacking interactions, different 4XPEA halogens also greatly impact molecular packing density. Crystallographic analysis of the interlayer packing density—defined as the ratio of the volume occupied by the spacer cations to the interlayer void volume³²—is maximized for $(4\text{CIPEA})_2\text{SnI}_4$ compared with the rest of the $(4\text{XPEA})_2\text{SnI}_4$ series (Fig. 1f). By contrast, the smaller molecular volume of 4FPEA is insufficient to offset the expanded interlayer, leading to a lower packing density. Meanwhile, the oversized bromine atom in 4BrPEA dramatically expands the interlayer and the π -stacking distances resulting from the increased steric hindrance, which also leads to lower packing density. Thus, the intermediate 4CIPEA cation effectively balances the trade-offs between molecular volume and interlayer expansion to enable the tightest interlayer packing. This comparison highlights interlayer packing density as a critical yet often overlooked design parameter in the development of layered perovskites. Importantly, each of the halogenated $(4\text{XPEA})_2\text{SnI}_4$ also shows stronger face-to-face interactions compared with unfunctionalized $(\text{PEA})_2\text{SnI}_4$ (purple bars in Fig. 1f), which also contributes to the rigidity of the organic interlayer. The combination of supramolecular interactions, packing density and hydrophobicity of the spacer cations each can influence the air and water stability of 2D THPs, which we investigate in the following section.

The tight packing enabled by 4CIPEA is also maintained in the $n = 2$ $(4\text{CIPEA})_2(\text{FA})\text{Sn}_2\text{I}_7$ structure (Fig. 1g and Supplementary Fig. 1). Notably, this structure also shows a perfectly undistorted out-of-plane Sn–I–Sn angle of 180.0° and low octahedral distortions (Supplementary Table 2), suggestive of an uncommon ‘balanced’ $n = 2$ quasi-2D perovskite structure²⁸. Such low structural distortions lead to enhanced metal halide orbital overlaps and photophysical characteristics^{27,28}, which, when coupled with the general appeal of quasi-2D perovskite passivation in PSCs^{45–47}, make 4CIPEA-based THPs highly appealing for tin PSCs.

Interlayer packing-dictated air and water stability

To evaluate the impacts of the halogenation of 4XPEA on the air stability of THPs, we tracked how $(4\text{XPEA})_2\text{SnI}_4$ (X = H, F, Cl, Br) maintained their crystallinity and PL emission as they aged under ambient air by monitoring the changes in their powder X-ray diffraction (PXRD) patterns and visible PL emission in bulk crystals. $(\text{PEA})_2\text{SnI}_4$ degraded rapidly under ambient conditions: new PXRD peaks corresponding to degradation products emerged after only 2 weeks, and the initially bright red PL emission (under temporary excitation by a UV flashlight) became dim after 1 week (Fig. 2a). In stark contrast, $(4\text{FPEA})_2\text{SnI}_4$, $(4\text{CIPEA})_2\text{SnI}_4$ and $(4\text{BrPEA})_2\text{SnI}_4$ showed much greater air stability over several weeks of continuous air exposure. $(4\text{FPEA})_2\text{SnI}_4$ and $(4\text{BrPEA})_2\text{SnI}_4$ showed similar signs of degradation to $(\text{PEA})_2\text{SnI}_4$ only after 8 weeks (Fig. 2b,d). Impressively, the new $(4\text{CIPEA})_2\text{SnI}_4$ phase showed no visible changes in its PXRD pattern and maintained bright PL emission over several months of ambient air exposure (Fig. 2c and additional images in Supplementary Fig. 3), showing that $(4\text{CIPEA})_2\text{SnI}_4$ undergoes negligible degradation after 16 weeks in air.

Beyond air stability, we assessed the resilience of various $(4\text{XPEA})_2\text{SnI}_4$ to water exposure (Supplementary Fig. 4). Compared with the $(\text{PEA})_2\text{SnI}_4$ and $(4\text{FPEA})_2\text{SnI}_4$ crystals that rapidly dissolved (or degraded) in water within 30 min, the $(4\text{CIPEA})_2\text{SnI}_4$ and $(4\text{BrPEA})_2\text{SnI}_4$ crystals showed minimal dissolution and maintained their initial black colour after 24 h in water. The differences between $(4\text{CIPEA})_2\text{SnI}_4$ and $(4\text{BrPEA})_2\text{SnI}_4$ only became more apparent after 6 days, when a greater proportion of $(4\text{CIPEA})_2\text{SnI}_4$ solid remained black compared with $(4\text{BrPEA})_2\text{SnI}_4$, which had been fully converted into a white product. The water stability trend is in general agreement with the packing density and air stability trend (that is, $4\text{CIPEA} > 4\text{BrPEA} > 4\text{FPEA} > \text{PEA}$), but is probably also further modulated by spacer cation solubility.

To better understand the mechanistic origins for the ultra-high air and water stability in $(4\text{CIPEA})_2\text{SnI}_4$, we used DFT to calculate O_2 and H_2O diffusion energy barriers across the organic interlayers in the $(4\text{XPEA})_2\text{SnI}_4$ (X = H, F, Cl, Br) structures. Two diffusion pathways were examined (Fig. 2e): a benzene-controlled path, in which the molecule passes between adjacent aromatic rings, and a halide-controlled path, in which it passes between the para-substituents of neighbouring 4XPEA cations. For O_2 , the lowest-barrier pathway depends strongly on the substituent (Fig. 2f,g). In $(\text{PEA})_2\text{SnI}_4$ and $(4\text{FPEA})_2\text{SnI}_4$, the halide-controlled path has the lower barrier (0.19 eV and 0.40 eV), indicating that H and F are too small to block this path effectively. By contrast, in $(4\text{CIPEA})_2\text{SnI}_4$ and $(4\text{BrPEA})_2\text{SnI}_4$, Cl and Br strongly hinder the halide-controlled path (0.94 eV and 0.96 eV), forcing O_2 to diffuse through the benzene-controlled path. Along this route, the tighter aromatic packing in $(4\text{CIPEA})_2\text{SnI}_4$ gives a barrier of 0.85 eV, markedly higher than that in $(4\text{BrPEA})_2\text{SnI}_4$ (0.65 eV). Differential charge density analysis of the transition-state configurations (Supplementary Fig. 5) indicates that these barriers are governed primarily by Pauli repulsion between O_2 and the surrounding organic framework. Calculations of H_2O diffusion barriers show the same ranking, with $(4\text{CIPEA})_2\text{SnI}_4$ again showing the highest minimum barrier (0.77 eV) (Supplementary Fig. 6). These results identify Cl as the optimal substituent in $(4\text{XPEA})_2\text{SnI}_4$, because it is large enough to block the halide-side path, yet not too large to expand the interlayer π – π stacking and lower the barrier along the benzene-controlled diffusion path.

To assess the kinetic consequences of these different energy barriers, we estimated relative diffusion rates using a thermally activated rate model (Supplementary Note 2). At 298 K, the predicted O_2 diffusion rate in $(4\text{CIPEA})_2\text{SnI}_4$ is lower than those in $(\text{PEA})_2\text{SnI}_4$, $(4\text{FPEA})_2\text{SnI}_4$ and $(4\text{BrPEA})_2\text{SnI}_4$ by ~11, 8 and 3 orders of magnitude, respectively (Fig. 2h). The corresponding H_2O diffusion rates are reduced by ~7, 6 and 1 orders of magnitude at 298 K (Supplementary Fig. 6b), consistent with the observed water stability (Supplementary Fig. 4). Together, these experimental and theoretical results show that strong supramolecular interactions and dense interlayer packing in $(4\text{CIPEA})_2\text{SnI}_4$ uniquely

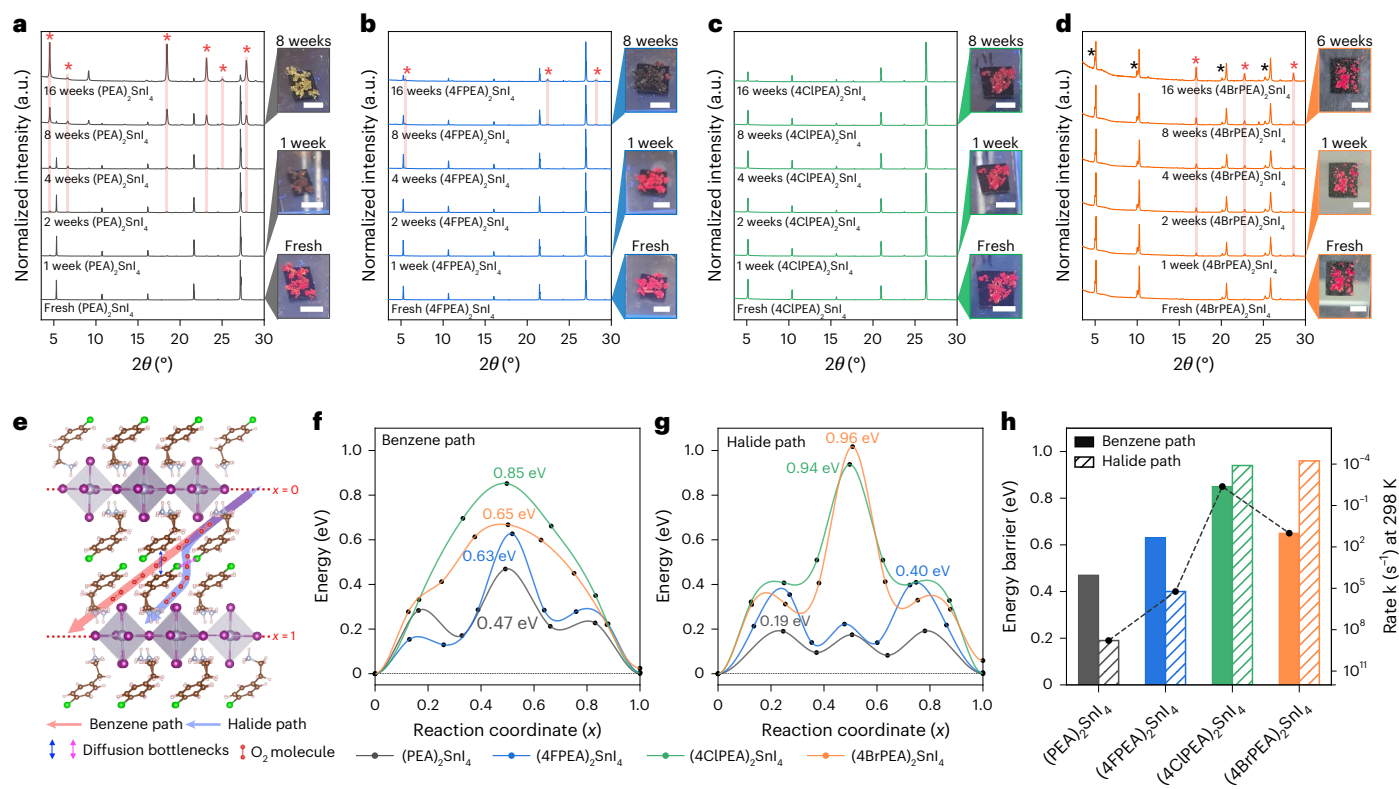


Fig. 2 | Air stability comparisons of (4XPEA)₂SnI₄ THPs. a–d, PXRD patterns for crushed crystals of (PEA)₂SnI₄ (a), (4FPEA)₂SnI₄ (b), (4CIPEA)₂SnI₄ (c) and (4BrPEA)₂SnI₄ (d) aged under ambient air (relative humidity, RH = 20–50%, 20.0 °C ± 2.0 °C) over 16 weeks under dark conditions (except for when they were briefly removed for periodic measurements). The red asterisks indicate new crystalline impurity peaks corresponding to the degradation of the 2D THP; black asterisks in the (4BrPEA)₂SnI₄ pattern correspond to 4BrPEAI salt coprecipitate. Insets show optical images of (4XPEA)₂SnI₄ crystals aged under the same conditions and photographed under temporary excitation by a 385-nm UV flashlight. Scale bars, 0.5 cm. Both PXRD and optical images suggest that (4CIPEA)₂SnI₄ shows negligible degradation after 16 weeks, but the other (4XPEA)₂SnI₄ shows various degrees of degradation. **e**, Schematic depicting the

two major DFT-calculated O₂ diffusion paths (benzene-controlled and halide-controlled) across the organic interlayer in the (4XPEA)₂SnI₄ structure. **f, g**, Calculated energy profiles of O₂ diffusion across the (4XPEA)₂SnI₄ interlayers through the benzene-controlled path (f) and halide-controlled path (g). **h**, Calculated energy barriers (left axis, linear) and corresponding hopping rates $k = \nu \exp(-E_a/k_B T)$ at 298 K (right axis, inverted log scale) for O₂ diffusion through the benzene-controlled and halide-controlled paths in (4XPEA)₂SnI₄. E_a is the activation energy barrier for a given diffusion path, k_B is the Boltzmann constant, T is the absolute temperature; an attempt frequency $\nu = 10^{12} \text{ s}^{-1}$ is assumed throughout. The dashed line connects the lower barrier (faster diffusion) path for each material.

block all major O₂ and H₂O diffusion pathways, thereby enabling its exceptional air and water stability. In principle, these packing density insights could also be leveraged to mitigate ion migration in alloyed perovskite compositions.

Influence of 4XPEA substituent on THP thin film growth

We next investigated the influence of 4XPEA cations on the morphology, crystallinity and extent of orientation of spin-coated thin films of the prototypical 3D THP, FASnI₃. Given the substantially improved air and water stability of (4CIPEA)₂SnI₄, our subsequent optimizations focused on the 4CIPEA spacer cation, in comparison with PEA and 4FPEA. During the initial ink optimizations using varying amounts of 4CIPEA, we found that films with 5% 4CIPEAI salt (mol% relative to FASnI₃) showed the strongest FASnI₃ XRD intensities (Supplementary Fig. 7). Thus, 5 mol% was chosen as the baseline for comparisons between these spacer cations. Scanning electron microscope images show that the inclusion of 4XPEA (X = H, F, Cl) in the THP inks clearly improves the film morphology of these FASnI₃ (+5% 4XPEAI) films (Fig. 3a–d). Compared with the purely 3D FASnI₃ film that possesses large pinholes and distinct grain boundaries (Fig. 3a), the +5% phenethylammonium iodide (PEAI) film shows a pinhole-free surface but with small grains and similarly sharp grain boundaries (Fig. 3b). Furthermore, both the +5% 4FPEAI

(Fig. 3c) and +5% 4CIPEAI (Fig. 3d) films show improved grain sizes, visibly smoother surfaces and obscured grain boundaries. The trends in film morphology are corroborated by their XRD patterns (Fig. 3e). The intensity of the (100) peak increases while the peak width decreases across the series (full width half maximum values of 0.117°, 0.086°, 0.083° and 0.079° for FASnI₃, +5% PEA, +5% 4FPEAI and +5% 4CIPEAI films, respectively), reflective of the marked improvements to grain size, quality, uniformity and thus overall film crystallinity with 4CIPEAI. Furthermore, the intensity of the (111) peak becomes suppressed with +5% 4FPEAI and 4CIPEAI, suggesting that these additives also enhance the (100) orientation of the FASnI₃ films.

Notably, when much greater concentrations of 4CIPEAI (10–50 mol%) were included in (4CIPEA)_xFA_{1-x}SnI₃ films, PXRD showed distinct mixtures consisting only of the $n = 2$ and 3D phases (Supplementary Fig. 8). This suggests that only (4CIPEA)₂(FA)SnI₇ templates the FASnI₃ films, which is preferable for controlled interfacial dynamics^{45,46}. Interestingly, this behaviour is unique to FA-based THPs, as analogous experiments with MASnI₃ films yielded mixtures of $n = 1, 2$ and 3D phases (Supplementary Fig. 9). To further confirm the templating behaviour of FASnI₃ with 4CIPEAI, we compared 2D grazing-incidence wide-angle X-ray scattering (GIWAXS) of the FASnI₃ + 5% 4XPEAI films (Fig. 3f, g and full results in Supplementary Fig. 10). The GIWAXS pattern for the +5% 4CIPEAI film showed distinct FASnI₃ reflections

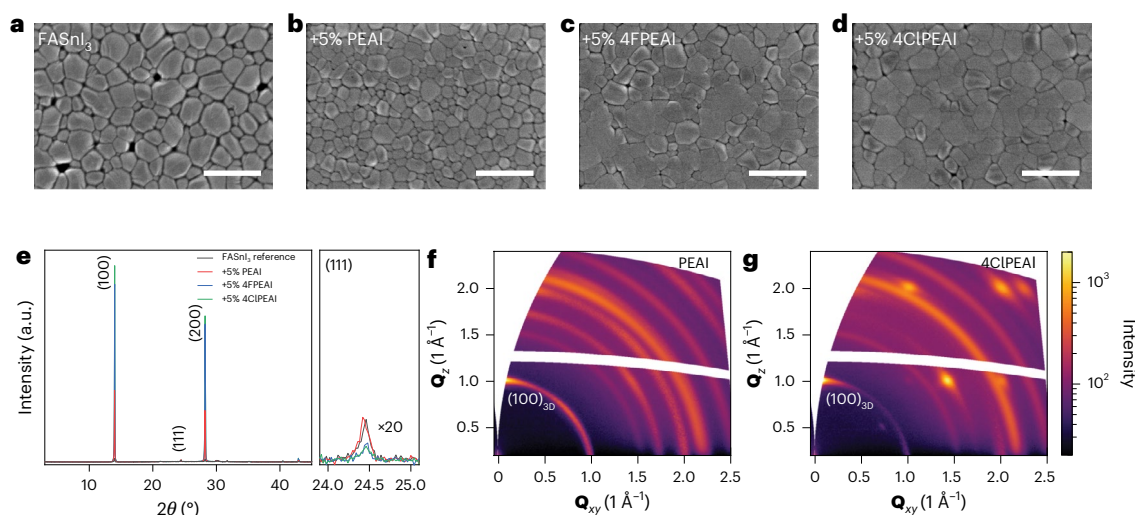


Fig. 3 | The effects of 4XPEA substituent on 2D/3D THP thin film growth. **a–d**, Scanning electron microscopy images of FASnI₃ (**a**), FASnI₃ + 5% PEAI (**b**), FASnI₃ + 5% 4FPEAI (**c**) and FASnI₃ + 5% 4CIPEAI (**d**) thin films grown on PEDOT:PSS-coated ITO (percentages are mol% relative to 0.8 M FASnI₃). Scale

bars, 2 μm . **e**, XRD patterns for the FASnI₃ (+5% 4XPEA) films. **f,g**, GIWAXS patterns of the FASnI₃ films with 5% PEAI (**f**) versus 5% 4CIPEAI (**g**). The colour bar shows diffraction intensity, while Q_{xy} and Q_z are the in-plane and out-of-plane scattering vectors, respectively.

compared with more isotropic Debye–Scherrer rings in the other compositions (Supplementary Fig. 10). These results confirm the superior film crystallinity and much stronger preferred orientation in films with 4CIPEA, probably owing to the enhanced π – π interactions in (4CIPEA)₂SnI₄ that could enable stronger templating^{35,40}. Time-of-flight secondary ion mass spectrometry (TOF-SIMS) shows that a major proportion of 4CIPEA resides at the buried interface of the film (Supplementary Fig. 12), which agrees with previous investigations of 2D/3D THP crystallization⁴⁰. Importantly, X-ray photoelectron spectroscopy (XPS) measurements on aged films revealed a smaller fraction of Sn⁴⁺ with the inclusion of 4CIPEA compared with the control FASnI₃ films (Supplementary Fig. 13), which shows the universal stability benefits offered by 4CIPEA, even in primarily 3D perovskite compositions.

To better understand the solution chemistry of the 4XPEA cations in the precursor inks, we also performed ¹H nuclear magnetic resonance (NMR) spectroscopy. Interestingly, in the presence of SnI₂ and formamidinium iodide (FAI), the chemical shifts of the aromatic protons of 4CIPEA become magnetically and electronically equivalent despite their chemical inequivalence (Supplementary Fig. 14). By contrast, the solutions containing PEAI or 4FPEAI showed minimal changes to aromatic proton chemical shifts and line profiles upon the addition of SnI₂ or FASnI₃. No evidence of chemical reactions, such as ring functionalization, was detected in the corresponding ¹³C and ¹¹⁹Sn NMR spectra (Supplementary Figs. 15 and 16). The convergence of the *ortho* and *meta* ¹H NMR chemical shifts of 4CIPEA in the presence of FAI and SnI₂ further suggests the presence of strong supramolecular π – π interactions^{42,43} and/or additional noncovalent interactions⁴⁸ that are maintained in solution under suitable conditions (Supplementary Fig. 17 and Note 3). Thus, the strong supramolecular interactions possible with 4CIPEA not only suppress oxidation in the solid state as discussed above, but may also govern crystallization of thin films through precursor solution phase interactions.

Stable narrow bandgap 2D/3D tin PSCs

We fabricated narrow bandgap 2D/3D tin PSCs incorporating 4CIPEA using the prototypical indium tin oxide (ITO)/poly(3, 4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS)/THP/indene-C₆₀ bisadduct (ICBA)/bathocuproine (BCP)/Ag architecture (Fig. 4a). For the active THP layer, we used a (CsI)_{0.05}(PEA_{0.15}FA_{0.85}SnI_{2.85}Br_{0.15})_{0.95} composition analogous to the current champion tin PSCs^{9,15},

but with an additional 5 mol% of 4XPEAI (relative to Sn) added into the precursor inks (see details in Methods). Figure 4b shows the current density–voltage (*J*–*V*) curves of the champion devices for each of the tested compositions, where control denotes (CsI)_{0.05}(PEA_{0.15}FA_{0.85}SnI_{2.85}Br_{0.15})_{0.95} without any additional additives, whereas (4X)PEAI denotes the same control composition with an additional 5% of the corresponding 4XPEAI salt (X = H, F, Cl). Steady improvements in the PCE were obtained across the spacer cation series, and the devices made using 5% 4CIPEAI achieved a maximum PCE of 16.19%. Such improvements are consistent with the superior crystallinity and templating behaviour from 4CIPEA discussed above, which are verified in the device compositions using PXRD (Supplementary Fig. 18). Statistical analysis across 14 devices for each composition shows that the trends in PCE are highly reproducible (Fig. 4c), with average PCE values of 10.34% $\pm$ 0.29% for the control, 12.53% $\pm$ 0.20% for PEAI, 14.16% $\pm$ 0.15% for 4FPEAI and 15.88% $\pm$ 0.13% for 4CIPEAI. Similar statistical plots for the open-circuit voltage *V*_{oc}, short-circuit current *J*_{sc} and fill factor confirm similarly reproducible improvements across the series (Supplementary Fig. 19 and Tables 4–7). Furthermore, the external quantum efficiency (EQE) spectra of these devices (Fig. 4d) yield integrated *J*_{sc} values that agree well with those found in the *J*–*V* measurements and revealed a consistent electronic bandgap of $\sim$ 1.44 eV (Supplementary Fig. 20), in a similar regime as other recent tin PSCs¹⁶.

Finally, we evaluated the long-term stability of the tin PSC devices using different 4XPEAI as they were stored under dry air (Fig. 4e). The duration for which 90% of the initial peak efficiency is retained (that is, the *T*₉₀ lifetime) improved across the series and maximized for 4CIPEAI. This follows the same stability trend shown by the (4XPEA)₂SnI₄ single crystals. Moreover, the champion 4CIPEAI device attained an impressive average *T*₉₅ surpassing 1,600 h in dry air, which is one of the highest ambient device stabilities for tin PSCs reported to date (see detailed comparison of recent reports in Supplementary Table 8). The operational stability of the tin PSCs is similarly prolonged, with the 4CIPEAI device retaining 80% of its initial efficiency after 1,000 h under continuous 1-sun illumination at 55 °C in air (Fig. 4f). Owing to the effective 2D materials design, unencapsulated devices with 4CIPEA also show the greatest ambient stability over the other 4XPEAI additives, even under high humidity (Supplementary Fig. 23). These results show how the use of 4CIPEA spacer cation, which forms ultrastable 2D THPs, can simultaneously improve both device efficiency

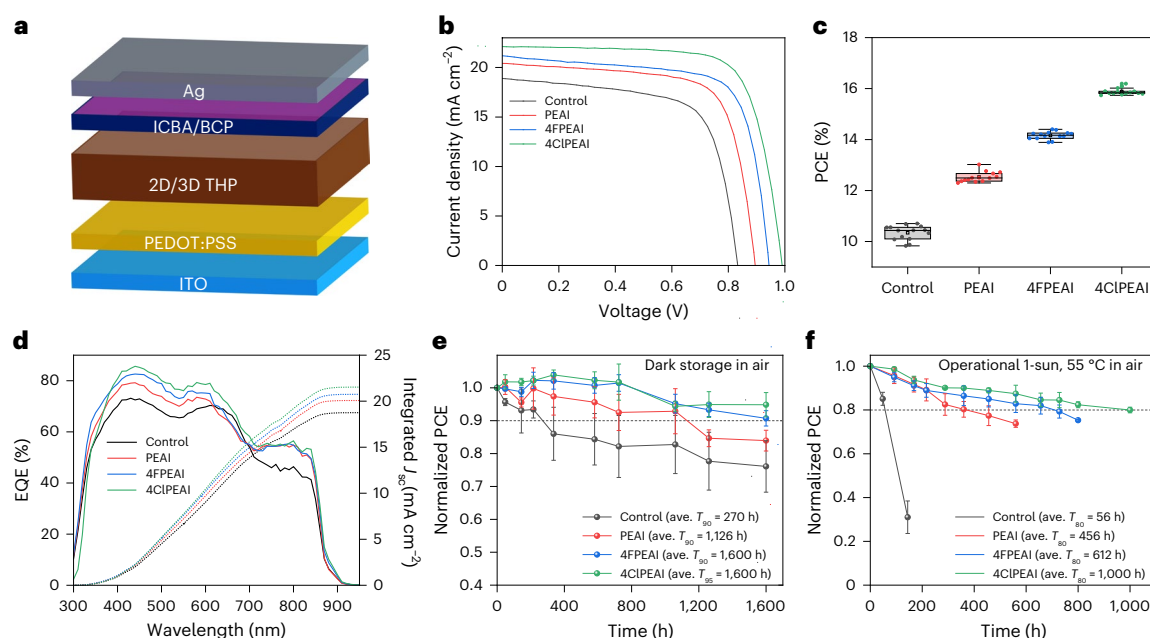


Fig. 4 | Performance and stability of narrow bandgap 2D/3D Sn PSCs based on 4XPEA. **a**, Device architecture of the 2D/3D tin PSC devices. **b**, J - V curves for the champion $(\text{CsI})_{0.05}(\text{PEA}_{0.15}\text{FA}_{0.85}\text{SnI}_{2.85}\text{Br}_{0.15})_{0.95}$ devices without 4XPEAI added (control) and with 5% of PEAI, 4FPEAI or 4CIPEAI, collected under simulated 1-sun AM1.5 illumination. **c**, PCE statistics based on 14 individual devices for each composition. The box plots contain the mean, median and 25th and 75th percentiles, with whiskers showing the 5th and 95th percentiles. **d**, EQE spectra (solid lines) for each composition with integrated J_{sc} (dash lines). **e**, Long-term stability of the tin PSCs with encapsulation under dry air storage.

Data are averaged over three devices per composition and normalized to their respective initial average (ave.) PCE values of 10.11%, 12.39%, 14.05% and 15.63% for the control, PEAI, 4FPEAI and 4CIPEAI devices, respectively. **f**, Operational stability of encapsulated tin PSCs under continuous 1-sun illumination at 55 °C in air. Data are averaged over three devices per composition and normalized to their respective initial average PCE values of 9.90%, 12.16%, 13.98% and 15.55% for the control, PEAI, 4FPEAI and 4CIPEAI devices, respectively. Error bars in **e** and **f** are shown as the standard deviation for each composition.

and stability to produce lead-free 2D/3D PSCs that are among the current state-of-the-art^{15,16}.

Outlook

In summary, we introduce a new design strategy for achieving ultrastable 2D THPs through spacer cation chlorination and develop highly stable and efficient 2D/3D tin PSCs. The intermediately sized chlorine atom in 4CIPEA spacer cation provides the best balance between increased molecular volume and limited interlayer expansion, leading to densely packed organic interlayers in $(4\text{CIPEA})_2\text{SnI}_4$. The strong supramolecular π -stacking interactions and tight interlayer packing greatly suppress O_2 and H_2O migration rates at room temperature by approximately 11 and 7 orders of magnitude compared with those in $(\text{PEA})_2\text{SnI}_4$. As a result, the 2D $(4\text{CIPEA})_2\text{SnI}_4$ retains its integrity for several months under ambient conditions, surpassing all known 2D THPs and related $(4\text{XPEA})_2\text{SnI}_4$. The incorporation of 4CIPEA also improves the morphology, crystallinity and orientation in deposited thin films of 3D THPs, probably owing to its strong π - π interactions and templating behaviour during crystallization. Leveraging these improved single crystal and film characteristics, we demonstrate highly efficient and stable 2D/3D tin PSCs with a champion PCE of 16.2% and an impressive T_{95} storage stability surpassing 1,600 h in dry air and T_{80} operational stability over 1,000 h at 55 °C in air. Our paper marks a major stride in overcoming the inherent air and moisture sensitivity of tin-based perovskites and further paves the way for their practical photovoltaic applications.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41563-026-02726-z>.

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Methods

Materials

For synthesizing single crystals, tin(II) chloride dihydrate (Sigma-Aldrich, 98% ACS reagent), 4CIPEA (Sigma-Aldrich, 98%), PEAI (GreatCell Solar), 4FPEA (TCI, >98.0%), 4BrPEA (Tokyo Chemical Industry, >98.0%), MA iodide (Sigma-Aldrich), FAI (GreatCell Solar), hydroiodic acid $\geq 47.0\%$ with $\leq 1.5\%$ hypophosphorous acid stabilizer (Sigma-Aldrich) and hypophosphorous acid 50% (Sigma-Aldrich) were used as provided. For synthesizing films, SnI_2 (Sigma-Aldrich, perovskite grade, 99.99%), SnF_2 (Sigma-Aldrich, 99%), FAI (GreatCell Solar), CsI (Sigma-Aldrich, AnhydroBeads, 99.999%), PEAI (GreatCell Solar), 4FPEAI (GreatCell Solar), 4CIPEAI (GreatCell Solar), ammonium thiocyanate (NH_4SCN ; TCI, >98%), ethylenediammonium diiodide (EDA_{I_2} ; Sigma-Aldrich, >98%), PEABr (Sigma-Aldrich, >98%), dimethylformamide (DMF; Sigma-Aldrich, anhydrous, 99.9%), dimethylsulfoxide (DMSO; Sigma-Aldrich, anhydrous, 99.9%), toluene (Sigma-Aldrich, anhydrous, 99.8%), chlorobenzene (Sigma-Aldrich, anhydrous, 99.8%), BCP (TCI), ICBA (1-Material), PEDOT:PSS (Heraeus Clevios) were used as provided.

2D tin iodide perovskite crystal growth

$(4\text{CIPEA})_2\text{SnI}_4$ was synthesized using 0.2 mmol $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ and 0.4 mmol 4CIPEA in 3.0 ml HI and 3.0 ml H_3PO_2 . $(4\text{CIPEA})_2(\text{FA})\text{SnI}_4$ was synthesized using 1.0 mmol $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$, 1.5 mmol FAI and 0.3 mmol 4CIPEA in 4.0 ml HI and 4.0 ml H_3PO_2 . $(4\text{CIPEA})_2(\text{MA})\text{SnI}_4$ was synthesized using 1.0 mmol $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$, 2.0 mmol MA iodide and 0.3 mmol 4CIPEA in 3.5 ml HI and 3.5 ml H_3PO_2 . $(\text{PEA})_2\text{SnI}_4$ was synthesized using 0.5 mmol $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ and 1.0 mmol PEAI in 1.5 ml HI and 1.5 ml H_3PO_2 . $(4\text{FPEA})_2\text{SnI}_4$ was synthesized using 0.2 mmol $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ and 0.4 mmol 4FPEA in 1.5 ml HI and 1.5 ml H_3PO_2 . $(4\text{BrPEA})_2\text{SnI}_4$ was synthesized using 0.25 mmol $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ and 0.1 mmol 4BrPEA in 1.5 ml HI and 1.5 ml H_3PO_2 . Solid precursors were added into vials with their corresponding amounts of 1:1 v/v HI and H_3PO_2 , followed by dropwise addition of the liquid amines (4CIPEA, 4FPEA, 4BrPEA). All solutions were heated to full dissolution at 120°C in an oil bath, then cooled by 10°C every 30 min (most samples) or slow-cooled over 48 h (for $(4\text{BrPEA})_2\text{SnI}_4$) to room temperature to yield large black crystals with plate morphology.

Preparation of perovskite precursor inks

All precursor inks were prepared in a nitrogen-gas filled glove box with <1.0 ppm O_2 and H_2O . For the FASnI_3 (+4XPEAI) inks, FAI, SnI_2 and SnF_2 were combined in a stoichiometric molar ratio of 1:1:0.1 and dissolved in 1 ml DMF/DMSO solvent (4:1 v/v) ratio to prepare a perovskite precursor solution with a molarity of 0.8 M FASnI_3 . Then, 5 mol% XPEAI was added into the perovskite solutions, which were stirred for 2 h and then filtered with 0.2 μm polytetrafluoroethylene (PTFE) syringe filters before use.

For preparing the precursor inks for fabricating the PSC devices, FAI, PEABr, NH_4SCN , SnI_2 and SnF_2 were combined in a stoichiometric molar ratio of 0.85:0.15:0.05:1:0.1 and dissolved in 1 ml DMF/DMSO solvent (4:1 v/v ratio) to prepare a perovskite precursor solution with a molarity of 0.9 M. Then, 5 mol% XPEAI was added into the perovskite solutions. After complete dissolution, a 1.0-M CsI stock solution in DMSO was mixed with the perovskite precursor solutions to achieve a 5-mol% Cs ratio. The precursor solutions were stirred for 2 h and filtered with 0.2 μm PTFE filters before use.

Sn perovskite thin film deposition and PSC fabrication

All tin perovskite films were deposited on PEDOT:PSS/ITO substrates unless noted otherwise. Prepatterned ITO substrates were ultrasonically cleaned sequentially in acetone and isopropanol baths for 30 min each, followed by ultraviolet–ozone treatment for 20 min. Filtered PEDOT:PSS (through 0.45 μm polyvinylidene fluoride, PVDF, filter) was spin-coated onto the ITO substrates at 5,000 rpm for 30 s, followed by immediate annealing at 150°C for 20 min. For depositing the

FASnI_3 (+5% 4XPEAI) films, the respective inks (described above) were spin-coated at 5,000 rpm for 30 s, with 800 μl of chlorobenzene dispensed 10 s into the spinning process, followed by annealing at 100°C for 10 min. For fabricating the tin PSC devices, the respective inks were spin-coated at 5,000 rpm for 30 s, with 500 μl of toluene dispensed 10 s into the spinning process, followed by annealing at 80°C for 10 min and subsequent surface treatment with EDA_{I_2} (1 mg ml^{-1} in chlorobenzene) at 80°C for 5 min. ICBA solution (18 mg ml^{-1} in chlorobenzene) was spin-coated onto the THP films at 1,000 rpm for 30 s, followed by annealing at 80°C for 10 min. Lastly, BCP (6 nm) and Ag (100 nm, masked) were sequentially deposited onto the films by thermal evaporation. The THP films and devices were kept in a N_2 -filled glove box with <1.0 ppm O_2 and H_2O for all manipulations unless noted otherwise.

XRD

PXRD patterns of crushed $(4\text{XPEA})_2\text{SnI}_4$ crystals were collected using a Bruker D8 Advance X-ray diffractometer, and PXRD of deposited THP films were collected using a Rigaku D-Max 2200; both were taken with $\text{Cu K}\alpha$ radiation in Bragg–Brentano geometry. Single-crystal XRD data collection was performed using a Bruker Quazar SMART diffractometer with $\text{Mo K}\alpha$ ($\lambda = 0.71073 \text{ \AA}$) radiation (operation at 50 kV and 0.6 mA). A successful solution by intrinsic phasing provided most nonhydrogen atoms from the E-map. The remaining nonhydrogen atoms were located with an alternating series of least-squares cycles and difference Fourier maps. All nonhydrogen atoms were refined with anisotropic displacement coefficients. All hydrogen atoms were included in the structure factor calculation at idealized positions and allowed to ride on the neighbouring atoms with relative isotropic displacement coefficients. The crystal structure parameters (bond angles, bond lengths, octahedral distortions, spacer cation volumes, and others) were analysed using OLEX2 crystallography software. Interlayer packing density was calculated as the ratio of the spacer cation volume to the total interlayer volume, using a previously outlined method³².

Electron microscope characterization

The morphologies of the prepared THP thin films were investigated using a Hitachi S-4800 scanning electron microscope with a 3.0 kV accelerating voltage.

GIWAXS characterization

GIWAXS measurements were performed on a Xenocs Xeuss 3.0 SAXS/WAXS instrument using $\text{Cu K}\alpha$ radiation ($\lambda = 1.54 \text{ \AA}$) and repeated on two sets of samples. The perovskite thin film samples were loaded into the sample chamber on a two-axis tilt stage (HUBER Diffraktionstechnik) and held under moderate vacuum ($P < 10 \text{ Pa}$) for the duration of the experiment. Diffraction images were collected on a Dectris Eiger 2IM detector (DECTRIS AG). The sample detector distance was 100 mm, and the incidence angle was set to 2° . The images reported here are an average of data collected at three positions on the sample for at least 10 min per acquisition.

TOF-SIMS

An ION-TOF TOF-SIMS.V TOF-SIMS was used for depth profiling of the perovskite using methods covered in detail in a previous report⁵⁰. Analysis was completed using a 3-lens 30-keV BiMn primary ion gun. High mass resolution depth profiles were completed with a 30-keV Bi_3^+ primary ion beam (0.8 pA pulsed beam current), and a $50 \times 50 \text{ }\mu\text{m}$ area was analysed with a 128:128 primary beam raster. Sputter depth profiling was accomplished with 1 keV caesium ion beam (6 nA sputter current).

XPS

XPS was performed on a Scienta-Omicron HiPP-3 system using monochromatic $\text{Al K}\alpha$ X-rays (1,486.6 eV), operated with a 900- μm spot size at 300 W. Scans were taken with pressure between 10^{-8} mbar and 5×10^{-8} mbar, using 200 eV pass energy and $1.5 \text{ mm} \times 30 \text{ mm}$ entrance

slit, which approximately corresponds to 0.75 eV energy resolution. Binding energy calibration had been performed on silver, where the binding energy for Ag 3d_{5/2} was set to 368.23 eV. Atomic percentages were calibrated using freshly cleaned polyethylene terephthalate standard. Processing of XPS data was performed using CasaXPS.

NMR

All NMR sample preparation was performed in a N₂-filled glove box. The perovskite reagents were dissolved in a 4:1 *v/v* ratio of DMF to DMSO-d₆ (Sigma-Aldrich, 99.5% deuterated). The majority of ¹H and ¹³C NMR measurements were carried out on a 400-MHz Bruker AVII-IHD Nanobay spectrometer equipped with a liquid nitrogen-cooled HX Prodigy cryoprobe using the following default acquisition parameters: for ¹H NMR—zg30 Bruker pulse sequence, 20 ppm spectral width, 64,000 acquired points, 16 scan averages and a 3-s recycle delay; for ¹³C NMR—zgpg30 Bruker pulse sequence, 240 ppm spectral width, 128,000 acquired points, 8,000 scan averages and a 2-s recycle delay. Additional ¹H NMR spectra were collected at a higher field for increased clarity of the coupling patterns. These spectra were obtained on a 600-MHz Bruker AV4 NEO NMR spectrometer equipped with a helium cooled, 5-mm triple resonance carbon inverse Cryoprobe (CPTCI), using a zg10 pulse sequence (zg30 Bruker sequence modified for a 10° tip angle), a 20-ppm spectral width, 64,000 acquired points (5.5 s acquisition time), 64 scan averages and a 1-s recycle delay. The ¹¹⁹Sn NMR spectra were taken on a 300-MHz Bruker AV4 NEO NMR Nanobay NMR spectrometer equipped with a 5-mm broadband fluorine observation (BBFO) probe. Typical acquisition parameters included a zg30 Bruker pulse sequence, a 667-ppm spectral width (75 kHz), 64,000 acquired points (0.4 s acquisition time), 128 scan averages and a 2-s recycle delay. The centre position of the spectrum was altered depending on the Sn environments expected. All spectra were locked and referenced to DMSO-d₆ at 2.50 ppm. All data were processed using MestreNova version 15.

Solar cell and optical characterization

The *J*–*V* curves were measured in a N₂-filled glove box using an Air Mass (AM) 1.5 G solar simulator (Oriel Sol3A Class AAA) and collected using a scan rate of 20 mV s^{−1} and a scan range of −0.2 V to 1.0 V. The intensity of the solar simulator was calibrated to 100 mW cm^{−2} using a reference silicon solar cell (Oriel, VLSI standards, certified by the National Laboratory of the Rockies Photovoltaic Performance Calibration Group). The device area was masked with a metal aperture to define an active area of 0.059 cm². UV–visible (UV–Vis) and absorbance measurements were collected using an Agilent Cary 4000i spectrophotometer. PL measurements were performed using a Horiba LABRAM HR Evolution Raman and PL spectrometer under 532 nm excitation. EQE was assessed on encapsulated devices in ambient air using a Newport Oriel IQE200 system, with a monochromatic beam focused on the device pixels and the light modulated by a chopper at 37 Hz.

Stability assessments were performed on encapsulated devices. To encapsulate the solar cells, in a N₂-filled glove box, UV-curable epoxy (Eversolar AB-341) was spread over the active perovskite area, a flat cover glass was placed on top and the seal was cured by gently pressing the glass while irradiating with <395 nm UV light for about 2 min. All solar cell devices used the same active compositions, surface treatments and ITO/PEDOT:PSS/THP/ICBA/BCP/Ag device architectures as those studied throughout Fig. 4. For the storage stability comparisons, devices were kept under dark conditions under dry air and periodically measured in a N₂-filled glove box. The PCE values for each composition were averaged over three devices and normalized to their respective initial average PCE values (10.11%, 12.39%, 14.05% and 15.63% for the control, PEAI, 4FPEAI and 4CIPEAI devices, respectively). On separate encapsulated devices, the light and operational stability was evaluated under a resistive load of 500 Ω, close to the maximum power point, while exposed to 1-sun illumination at 55 °C ± 2 °C in air with a relative

humidity of 30–50%. After defined intervals, the resistive load was disconnected and *J*–*V* measurements were collected using a Keithley 2450 source meter; the resulting curves were analysed to extract the relevant parameters. The PCE values for each composition were averaged over three devices and normalized to their initial average PCE values (9.90%, 12.16%, 13.98% and 15.55% for the control, PEAI, 4FPEAI and 4CIPEAI devices, respectively).

DFT calculations

All DFT calculations were carried out using the VASP code with projector augmented-wave pseudopotentials^{51,52}. A plane-wave cut-off of 600 eV and *Γ*-point sampling were adopted. The exchange–correlation interactions were treated within the generalized gradient approximation of the Perdew–Burke–Ernzerhof functional⁵³, while van der Waals forces were included using Grimme's D3 correction⁵⁴. Spin polarization was considered to accurately describe the triplet ground state of the O₂ molecule. The simulation supercells of (4XPEA)₂SnI₄ (X = H, F, Cl, Br) consisted of two SnI₄ monolayers with a total of 376 atoms. Diffusion pathways of O₂ and H₂O were investigated using the nudged elastic band method, as implemented in VASP via the VTST-Tools package⁵⁵. At least five intermediate images were used along each path, and all residual atomic forces were converged to below 0.02 eV Å^{−1} to obtain reliable energy profiles and migration barriers of O₂ and H₂O.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Cambridge Crystallographic Data Centre (CCDC) deposition numbers 2496775–2496777 and 2538309 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the CCDC available at www.ccdc.cam.ac.uk/data_request/cif. All data needed to evaluate the conclusions of the study are available in the Article or Supplementary Information.

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Author contributions

C.T.T., L.C., K.Z. and S.J. conceived the idea and designed the experiments. C.T.T. synthesized the 2D tin perovskites and performed the powder and single-crystal XRD studies. C.T.T. and F.Y. characterized the surface morphologies of the films. L.C., C.T.T. and F.Y. fabricated the tin PSC devices. L.C. conducted the EQE, *J*–*V*

and long-term device stability measurements. J.X. performed the DFT calculations and analysis, under the guidance of X.W. and Y.Y., which provided theoretical insights. N.J.W. and Z.Z. conducted the GIWAXS measurements and analysis under the guidance of M.F.T. M.T. performed the absorbance characterizations. T.L. and L.C. performed the UV–vis measurements. S.P.H. performed the TOF-SIMS measurements. J.Y.Y. performed the XPS measurements and analysis. J.Y.Y., R.A.K., B.A. and C.T.T. performed the NMR analyses. S.J., K.Z., Y.Y. and M.C.B. supervised the project. C.T.T., L.C., J.X., Y.Y., K.Z. and S.J. wrote the paper. All authors discussed the results and contributed to the revisions of the paper.

Competing interests

C.T.T., L.C., K.Z. and S.J. have filed a patent with the US Patent and Trademark Office (application # US64/048,398) based on this work. The other authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells

☒ Yes
☐ No

The active area of the tested solar cells is 0.059 cm².

Explain why this information is not reported/not relevant.

Method used to determine the device area

☒ Yes
☐ No

A metal aperture to define an active area of 0.059 cm² was used during J-V scans.

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction

☒ Yes
☐ No

Provided in this manuscript.

Voltage scan conditions

☒ Yes
☐ No

Provided in this manuscript.

Explain why this information is not reported/not relevant.

Test environment

☒ Yes
☐ No

Provided in this manuscript.

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization

☐ Yes
☒ No

Provide a description of the protocol.

No preconditioning was used in this work.

Stability of the J-V characteristic

☐ Yes
☒ No

Provide a description of the method used. The stability of the J-V characteristic can be verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 5 for details.

The software for J-V measurements was not able to measure the stabilized PCE.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization

☒ Yes
☐ No

Very minor hysteresis was observed for devices.

Explain why this information is not reported/not relevant.

Related experimental data

☒ Yes
☐ No

J-V curves under reverse and forward voltage scans were provided.

Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)

☒ Yes
☐ No

EQE spectra were provided.

Explain why this information is not reported/not relevant.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator

☒ Yes
☐ No

The integrated J_{sc} from EQE measurements were consistent with the J_{sc} from J-V scans.

Explain why this information is not reported/not relevant.

For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☐ No	Provide a description of the measurement conditions. Explain why this information is not reported/not relevant.
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	Provided in this manuscript. Explain why this information is not reported/not relevant.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	The reference cells were calibrated by NREL. Explain why this information is not reported/not relevant.
Calculation of spectral mismatch between the reference cell and the devices under test	☐ Yes ☒ No	Provide a value of the spectral mismatch and/or a description of how it has been taken into account in the measurements. Since the illumination spectrum was well aligned with the Si reference cell, spectral mismatch between the reference and the devices under test was not calculated.
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	A metal aperture mask with the area of 0.059cm² was used for testing. Explain why this information is not reported/not relevant.
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	Report the difference in the short-circuit current density values measured with the mask and aperture area. All J-V curves were measured with a metal aperture mask.
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☐ Yes ☒ No	Identify the independent certification laboratory. We did not get a chance to certify our solar cells.
A copy of any certificate(s)	☐ Yes ☒ No	Certificate copies should be provided in the Supplementary information. Please state the supplementary item number. We did not provide a certificate.
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	Provided in this manuscript. Explain why this information is not reported/not relevant.
Statistical analysis of the device performance	☒ Yes ☐ No	Provided in this manuscript. Explain why this information is not reported/not relevant.
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	Provided in this manuscript. Explain why this information is not reported/not relevant.